

EMPLOYMENT

Stanford University
Postdoctoral Scholar

Jul. 2026 – present

EDUCATION

University of Michigan
Ph.D. in Physics

Sept. 2020 – Jun. 2026

University of Science and Technology of China
B.S. in Physics

Sept. 2015 – Jul. 2019

RESEARCH INTERESTS

Computational many-body physics; strongly correlated systems; non-equilibrium dynamics; advanced numerical methods.

PUBLICATIONS

- **“Compact and Stable Representation of Real-Frequency Spectral Functions for Machine Learning”**,
by Xinyang Dong, Ido Zemach, **Lei Zhang**, Guy Cohen and Emanuel Gull,
available on 13 July 2026 in [arXiv:2607.11190](#).
- **“Hybrid Hamiltonian-diagrammatic quantum impurity solver”**,
by Yang Yu, Gaurav Harsha, **Lei Zhang**, Agnieszka Jażdżewska, Dominika Zgid, Xinyang Dong and Emanuel Gull,
available on 9 June 2026 in [arXiv:2606.11095](#).
- **“Multi-orbital dynamical mean-field theory with a complex-time solver”**,
by Yang Yu, **Lei Zhang**, Emanuel Gull, Xiaodong Cao and Xinyang Dong,
published on 11 May 2026 in [Phys. Rev. Research 8, 023142 \(2026\)](#).
- **“Validity and Limits of Low Order Hybridization Expansion Approaches for Multi-Orbital Systems”**,
by Dolev Goldberger, Ido Zemach, **Lei Zhang**, Yang Yu, Emanuel Gull, Guy Cohen and André Erpenbeck,
available on 4 May 2026 in [arXiv:2605.02228](#).
- **“Compact representation and long-time extrapolation of real-time data for quantum systems using the ESPRIT algorithm”**,
by André Erpenbeck, Yuanran Zhu, Yang Yu, **Lei Zhang**, Richard Gerum, Olga Goulko, Chao Yang, Guy Cohen and Emanuel Gull,
published on 16 March 2026 in [Phys. Rev. B 113, 115129 \(2026\)](#).
- **“Minimal pole representation for spectral functions”**,
by **Lei Zhang**, André Erpenbeck, Yang Yu and Emanuel Gull,
published on 3 June 2025 in [J. Chem. Phys. 162, 214111 \(2025\)](#).
- **“Minimal pole representation and analytic continuation of matrix-valued correlation functions”**,
by **Lei Zhang**, Yang Yu and Emanuel Gull,
published on 12 December 2024 in [Phys. Rev. B 110, 235131 \(2024\)](#) [Editors’ Suggestion].

- **“Green/WeakCoupling: Implementation of fully self-consistent finite-temperature many-body perturbation theory for molecules and solids”**,
by Sergei Isakov, Chia-Nan Yeh, Pavel Pokhilko, Yang Yu, **Lei Zhang**, Gaurav Harsha, Vibin Abraham, Ming Wen, Munkhorgil Wang, Jacob Adamski, Tianran Chen, Emanuel Gull and Dominika Zgid, published in [Comput. Phys. Commun. 306, 109380 \(2025\)](#).
- **“Steady-state properties of multi-orbital systems using quantum Monte Carlo”**,
by Andre Erpenbeck, Thomas Blommel, **Lei Zhang**, Wei-Ting Lin, Guy Cohen and Emanuel Gull, published on 4 September 2024 in [J. Chem. Phys. 161, 094104 \(2024\)](#).
- **“Feynman diagrammatics based on discrete pole representations: A path to renormalized perturbation theories”**,
by Daria Gazizova, **Lei Zhang**, Emanuel Gull and J. P. F. LeBlanc, published on 27 August 2024 in [Phys. Rev. B 110, 075158 \(2024\)](#).
- **“Minimal Pole Representation and Controlled Analytic Continuation of Matsubara Response Functions”**,
by **Lei Zhang** and Emanuel Gull, published on 24 July 2024 in [Phys. Rev. B 110, 035154 \(2024\)](#).
- **“Tensor train continuous time solver for quantum impurity models”**,
by A. Erpenbeck, W.-T. Lin, T. Blommel, **L. Zhang**, S. Isakov, L. Bernheimer, Y. Núñez-Fernández, G. Cohen, O. Parcollet, X. Waintal and E. Gull, published on 26 June 2023 in [Phys. Rev. B 107, 245135 \(2023\)](#).
- **“Graphical Representations and Worm Algorithms for the $O(N)$ Spin Model”**,
by Longxiang Liu*, **Lei Zhang***, Xiaojun Tan and Youjin Deng, published on 10 November 2023 in [Commun. Theor. Phys. 75, 115702 \(2023\)](#),
(*: equal contribution).
- **“Loop-Cluster Coupling and Algorithm for Classical Statistical Models”**,
by **Lei Zhang**, Manon Michel, Eren M. Elçi and Youjin Deng, published on 12 November 2020 in [Phys. Rev. Lett. 125, 200603 \(2020\)](#).

PRESENTATIONS

Oral Presentations

- Invited talk at SIAM CSE27, Pittsburgh, PA, 2027
- Invited talk at Purdue University, West Lafayette, IN, 2026
- APS Global Physics Summit, Denver, CO, USA, 2026
- APS Global Physics Summit, Anaheim, CA, USA, 2025
- APS March Meeting, Chicago, IL, USA, 2022
- APS March Meeting (Online), 2021

Poster Presentations

- Simons Foundation Superconductivity Summer School, Minneapolis, MN, USA, 2025
- Electronic Structure Workshop, El Paso, TX, USA, 2025
- School on Electron Correlations and Topology, Houston, TX, USA, 2025
- MQC Entanglement Conference, Lansing, MI, USA, 2024
- Autumn School on Correlated Electrons, Jülich, Germany, 2023

SELECTED AWARDS

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|---|-------------|
| • Rackham One-Term Dissertation Fellowship | 2025 |
| • MICDE Fellowship | 2024 |
| • Outstanding Graduate of Anhui Province | 2019 |
| • Outstanding Graduate of USTC | 2019 |
| • Special Scholarship of SINANO, Chinese Academy of Sciences | 2018 |
| • National Encouragement Scholarship | 2017 |
| • First Prize in National College Students' Mathematics Competition | 2016 & 2017 |
| • Yan Jici Scholarship, Chinese Academy of Sciences | 2015 – 2019 |

TEACHING EXPERIENCE

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|---|---------------------------|
| • Physics 506 (Electromagnetism II), graduate course | Winter 2022 & Winter 2023 |
| • Physics 505 (Electromagnetism I), graduate course | Fall 2022 |
| • Physics 151 (Life Sciences Lab I), undergraduate course | Fall 2021 |
| • Physics 136 (Life Sciences Lab I), undergraduate course | Fall 2020 & Winter 2021 |

TECHNICAL SKILLS

Programming Languages: Python, C++, Fortran, MATLAB, Julia, Java, Mathematica

Libraries and Tools: NumPy, Matplotlib, Mpmath, Eigen, CMake, Git, $\LaTeX$, TikZ, Gnuplot